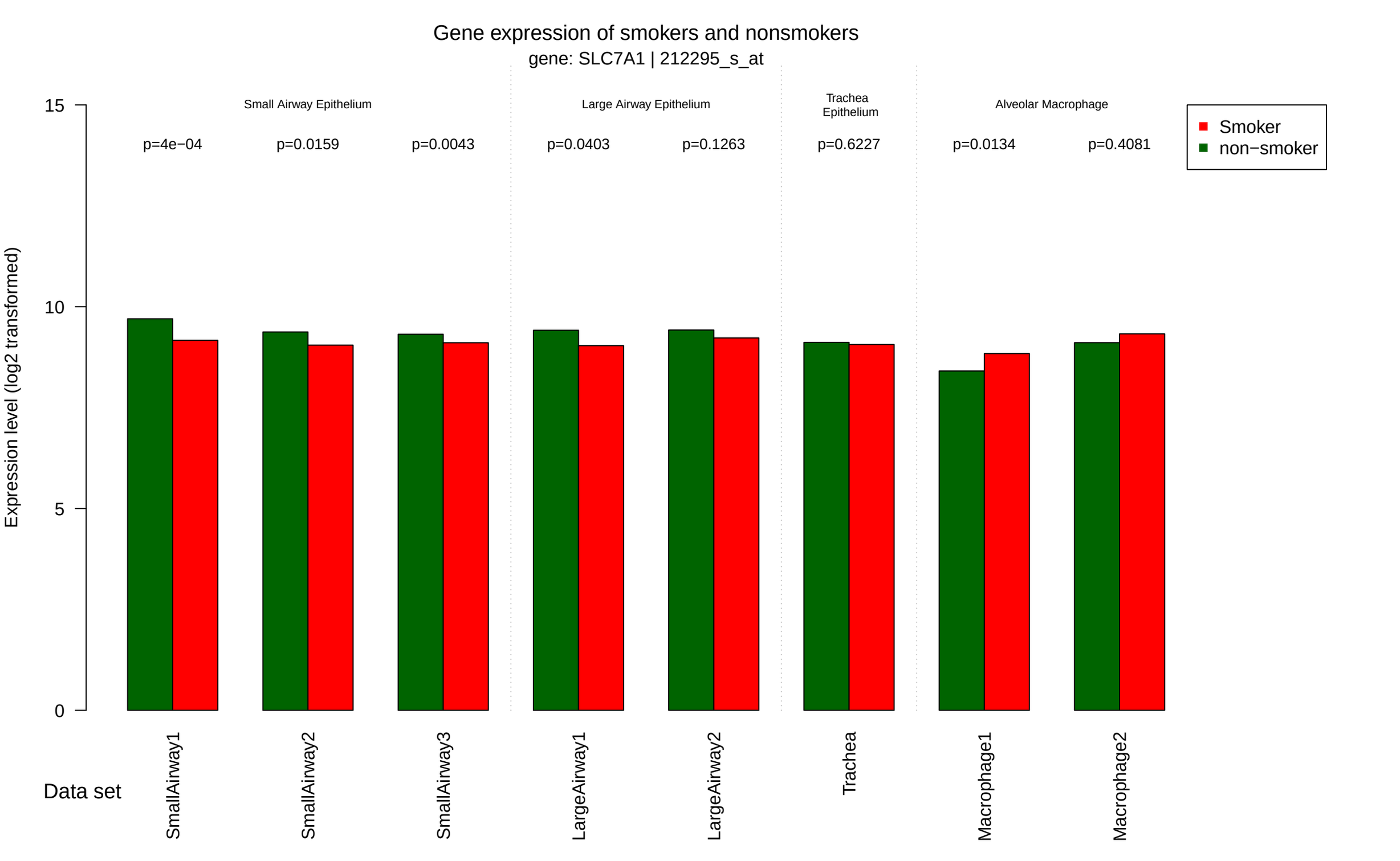
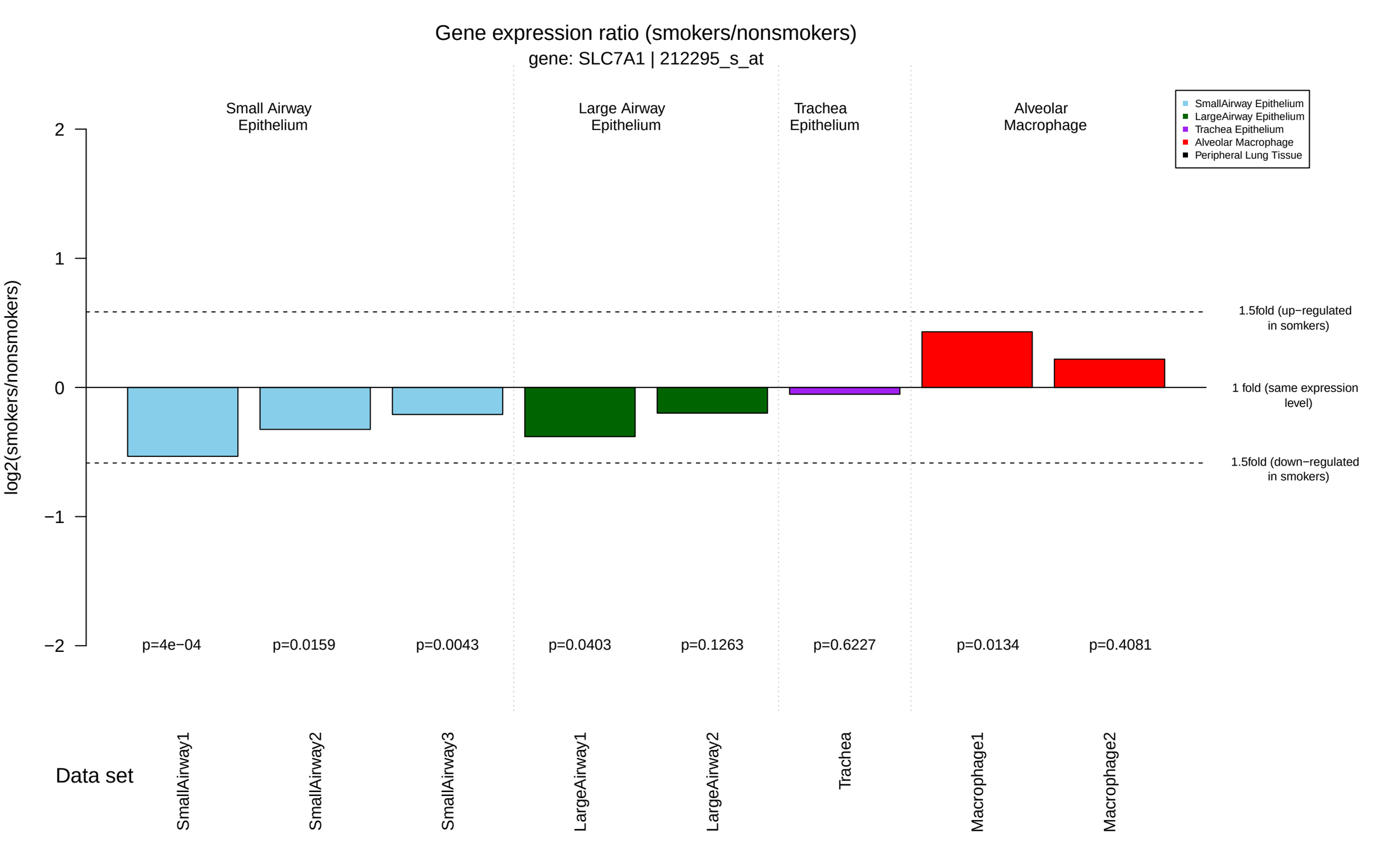
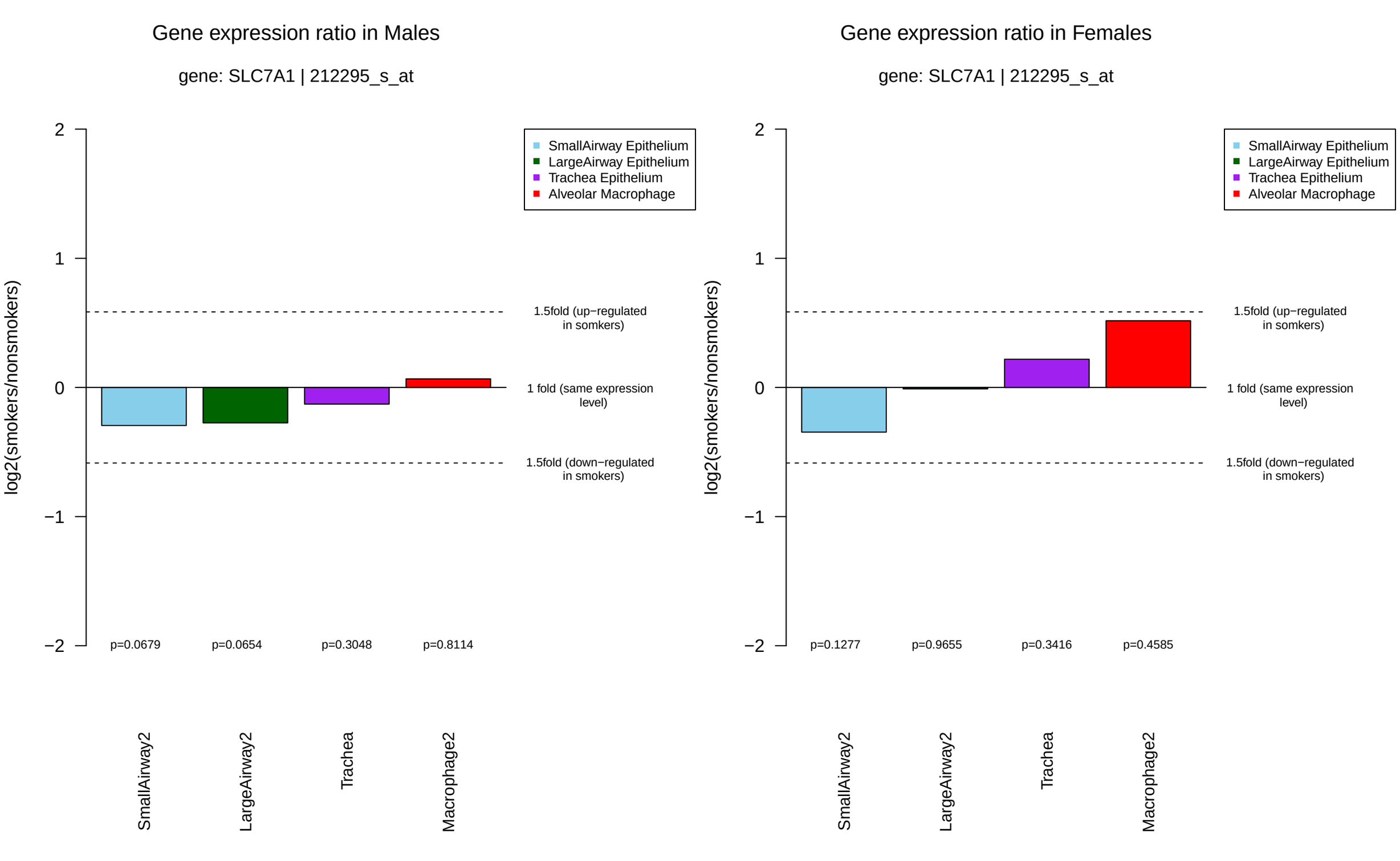
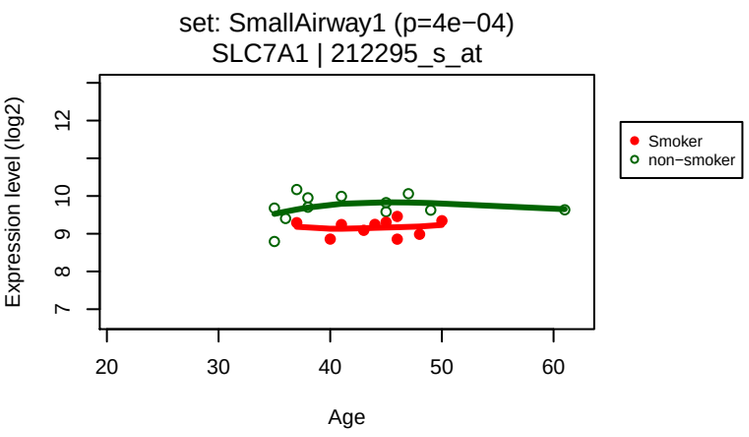
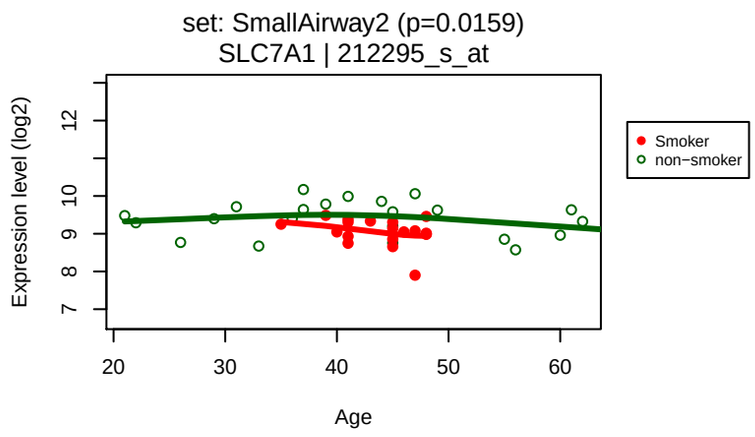
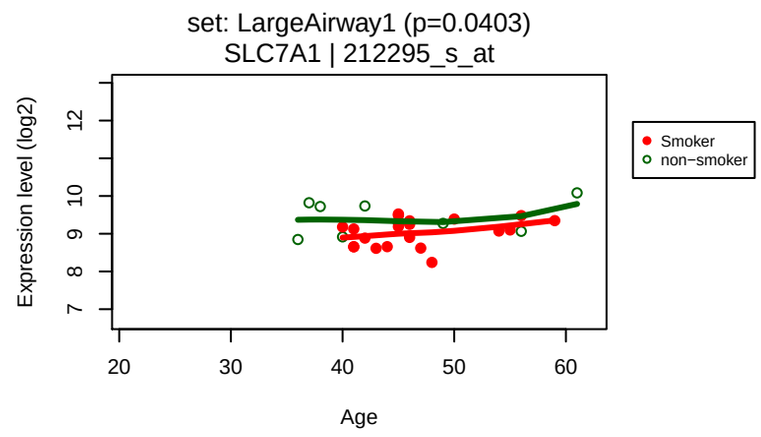

Gene expression of smokers and nonsmokers
gene: SLC7A1 | 212295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=4e−04
p=0.0159
p=0.0043
p=0.0403
p=0.1263
p=0.6227
p=0.0134
p=0.4081
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

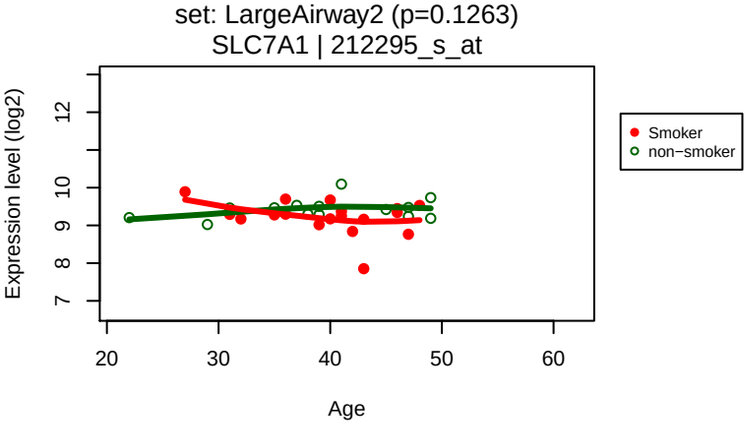

Gene expression ratio (smokers/nonsmokers)
gene: SLC7A1 | 212295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=4e−04
p=0.0159
p=0.0043
p=0.0403
p=0.1263
p=0.6227
p=0.0134
p=0.4081
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

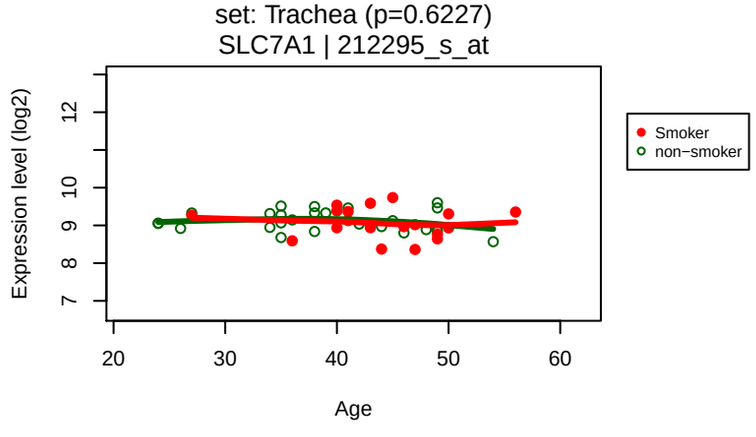

Gene expression ratio in Males
gene: SLC7A1 | 212295_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0679
p=0.0654
p=0.3048
p=0.8114
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC7A1 | 212295_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1277
p=0.9655
p=0.3416
p=0.4585
SmallAirway2
LargeAirway2
Trachea
Macrophage2

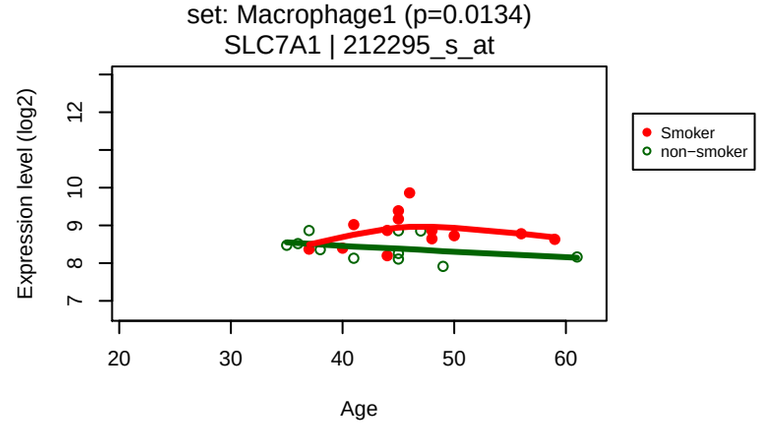

set: SmallAirway1 (p=4e−04)
SLC7A1 | 212295_s_at
Expression level (log2)
Age
Smoker
non−smoker

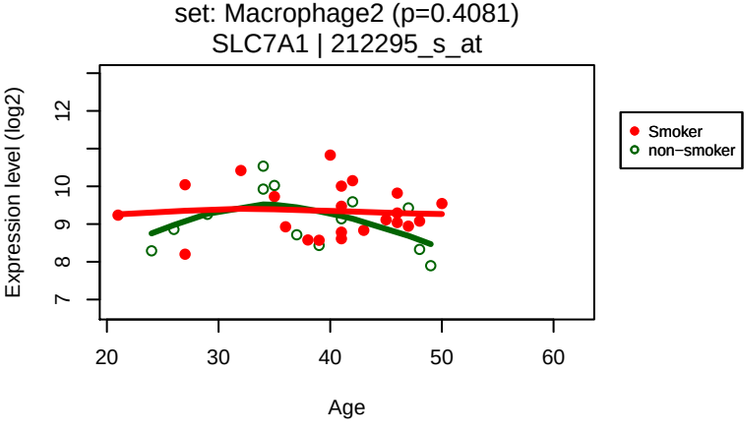

set: SmallAirway2 (p=0.0159)
SLC7A1 | 212295_s_at
Expression level (log2)
Age
Smoker
non−smoker

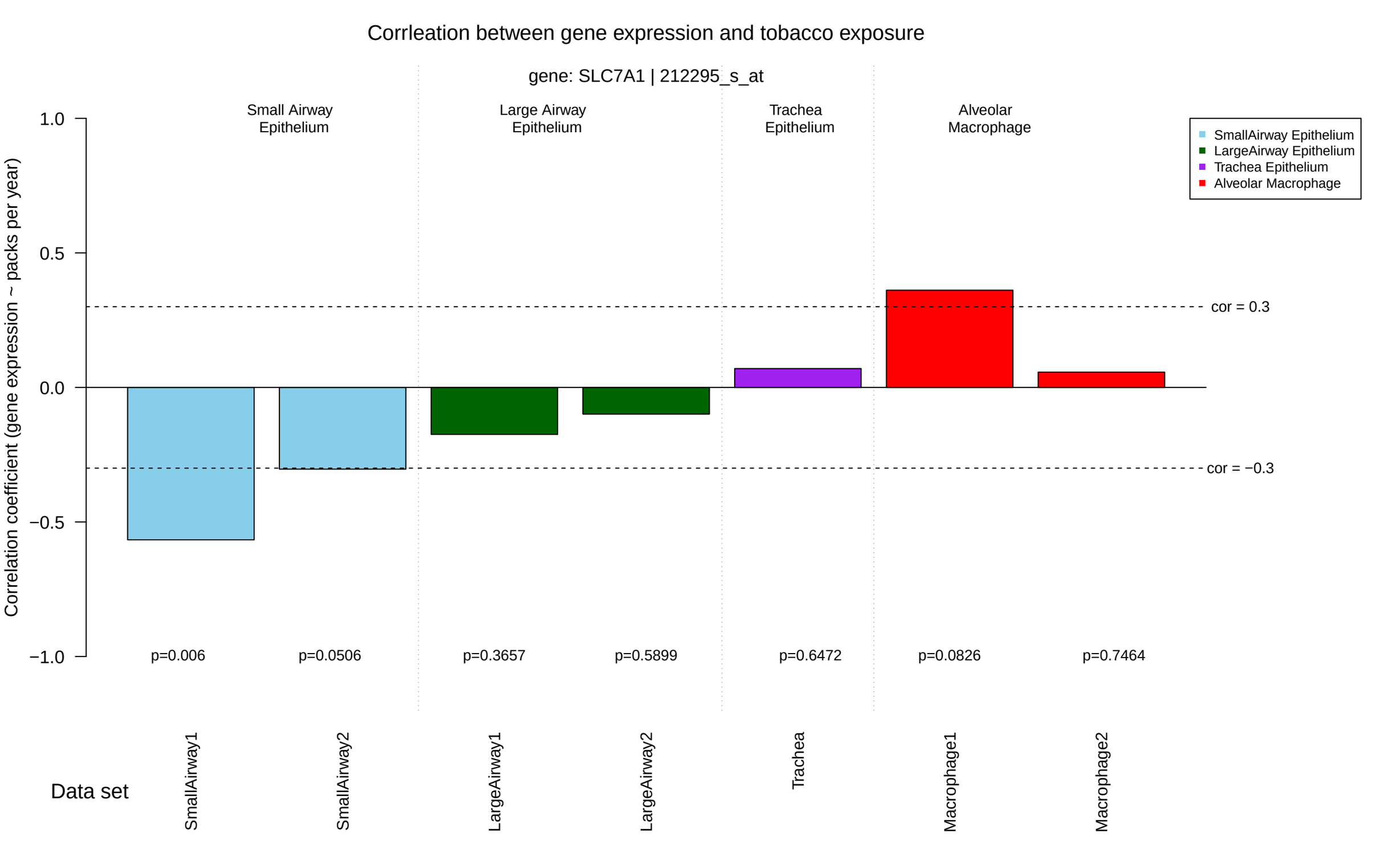

set: LargeAirway1 (p=0.0403)
SLC7A1 | 212295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1263)
SLC7A1 | 212295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6227)
SLC7A1 | 212295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0134)
SLC7A1 | 212295_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4081)
SLC7A1 | 212295_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC7A1 | 212295_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.006
p=0.0506
p=0.3657
p=0.5899
p=0.6472
p=0.0826
p=0.7464
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2